Wednesday Lab
16 Oct 2024

Q1. Write a program to find the smallest and largest digit in a number using functions.
Implement one function to extract the digits and another function to compare them.
Example:
Input: Enter a number: 4987
Output: Smallest digit: 4, Largest digit: 9

Q2. write a program that takes an array of size n from the user, prints it, and then cyclically rotates it clockwise m times as per the user's input
Example
Enter the size of the array: 5
Enter the elements of the array: 1 2 3 4 5
Original Array: 1 2 3 4 5
Enter the number of times to rotate the array: 2
Array after rotating 2 times: 4 5 1 2 3

Q3 Write a program that takes an array of size n from the user to sort the array in ascending order and return the sorted array. The sorting must be implemented without using any built-in sorting functions or libraries.

Q4. Write a program that takes an unsorted array of size n, and finds and returns any **peak element**.
Note: A peak element is an element greater than or equal to its neighbors. For the corner elements of the array, we only need to consider the one neighbour that exists. If the array is strictly increasing or decreasing, the peak will simply be the maximum value at the appropriate end of the array.

Example:

> **Input:** arr[]= {5, 10, 20, 15}
> **Output:** 20
> **Explanation:** The element 20 has neighbors 10 and 15, both of them are less than 20.
>
> **Input:** arr[] = {10, 20, 15, 2, 23, 90, 90}
> **Output:** 20 or 90
> **Explanation:** The element 20 has neighbors 10 and 15, both of them are less than 20, similarly 90 has neighbors 23 and 90.
>
> **Input:** arr[] = [1, 1, 1]
> **Output :** 1
> **Explanation:** All elements are peak in the given array, we can return any of them